

Annotated Bibliography

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References

- [1] Iker Pastor Izaskun Oregi Aitor Morais, Eneko Osaba. Comparative analysis of classical and quantum-inspired solvers: A preliminary study on the weighted max-cut problem. GECCO '25 Companion: Proceedings of the Genetic and Evolutionary Computation Conference Companion Pages 2449 - 2457, 2025.

This study compares three paradigms for solving the Weighted Max-Cut problem: a Genetic Algorithm, a Graph Neural Network, and a tensor network approach using Density Matrix Renormalization Group. Evaluating graphs from 10 to 250 nodes, it analyzes solution quality, runtime, and memory usage. Results show trade-offs in scalability, efficiency, and approximation performance.

- [2] Andrew J. Daley, Immanuel Bloch, Christian Kokail, Sean Flannigan, Nathan Pearson, Matthias Troyer, and Peter Zoller. Practical quantum advantage in quantum simulation. 2022.
- [3] Amr Magdy. Quantum optimization realized: Quantum annealing for combinatorial spatial problems in practice. Q-Spatial '25: Proceedings of the 1st ACM SIGSPATIAL International Workshop on Quantum Computing for Spatial Data Systems and Applications, 2026.

This manuscript summarizes D-Wave's keynote on quantum annealing as a practical approach to combinatorial optimization. It highlights D-Wave's QPUs for solving QUBO models and hybrid quantum-classical systems. Applications across academia and industry demonstrate that quantum annealing has matured into a production-ready technology delivering real-world optimization value.

- [4] Francesco Pio Marino Simone Faro. Scaling grover's search for large solution spaces. HPDC '25: Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing, 2025.

This work improves Grover's algorithm for cases where valid solutions exceed half the search space, a regime where performance typically degrades. By virtually expanding the domain with auxiliary qubits and modifying the oracle, the

method reduces effective solution density without increasing circuit depth. Simulations confirm improved reliability while preserving Grover’s optimal asymptotic complexity.

- [5] Juan Ungredda and Juergen Branke. Bayesian optimisation for constrained problems. *ACM Transactions on Modeling and Computer Simulation*, Volume 34, Issue 2, 2024.

The paper introduces a constrained variant of the Knowledge Gradient acquisition function for Bayesian optimisation to handle expensive black-box problems with constraints. It uses a Gaussian process surrogate to sequentially select evaluations, empirically outperforming four state-of-the-art constrained BO methods, and proves convergence to the true optimum in the infinite budget limit.

- [6] Thomas G. Wong. *Introduction to Classical and Quantum Computing*. Self-published, 2019.
- [7] Leo Zhou, Sheng-Tao Wang, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin. Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices. 2020.